

USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LAB #7 (Lesson 23) - Antenna Pattern Lab
Lab Sheet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 3.6: I can distinguish between array factor and true antenna pattern and account for element pattern effects.

This sheet is the turn-in document for the Lesson 23 lab. Report every lobe amplitude relative to the main-lobe peak, in dBc, and state the uncertainty next to any number read off a hand-walked trace. The lab is a team assignment, so one sheet per team.

- 1. The annotated trace.** Attach or sketch one screenshot of your best broadside run from the Tracking tab, with the main lobe and both first sidelobes labeled with their dBc values.

- 2. The amplitude comparison table.** Three lobes from the broadside run, each as an amplitude relative to the main-lobe peak. The mechanical column is your hand-walked trace, the electrical column is the Rectangular-tab sweep of step 4, and the calculated column comes from theory.

Lobe	Mechanical (dBc)	Electrical sweep (dBc)	Calculated (dBc)	Mechanical – electrical
Main lobe				
First sidelobe, left				
First sidelobe, right				

3. Written answers. Three or four sentences each.

- (a) Why do the lobe amplitudes survive a sloppy rotation while the lobe angles do not? Refer to what the time axis does and does not record.

- (b) Which of the four artifacts in Part 4 does an anechoic chamber remove, and which of them would still be present in a chamber measurement?

Documentation: _____